

After class

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About 25 minutes, after the lecture. Tick one answer for each multiple-choice question, show your working for each calculation, then check yourself with the worked answers at the end.

This activity checks all six outcomes

- EM-LO1** Define energy and heat, and convert between joules, calories and nutritional Calories
- EM-LO2** Convert between temperature scales and calculate heat gained or lost using $q = mc\Delta T$
- EM-LO3** Describe phase changes and calculate the energy involved using heats of fusion and vaporisation
- EM-LO4** Interpret the heating curve of water
- EM-LO5** Define bond energy, estimate ΔH from bond energies, and classify reactions as endothermic or exothermic
- EM-LO6** Relate energy changes to biochemical reactions such as glucose oxidation and ATP hydrolysis

1 · Concept check

EM-LO1

Q1. Which is the largest amount of energy?

- | | |
|-----------------------------------|------------------------------------|
| ☐ A 1 kJ | ☐ B 1 kcal |
| ☐ C 1000 J | ☐ D 100 cal |

EM-LO2

Q2. A laboratory keeps drug stability samples in a large water bath to hold them at a steady 37 °C. Why water?

- | | |
|--|--|
| ☐ A Water has a low specific heat, so it heats up quickly | ☐ B Water has a high specific heat, so its temperature changes slowly |
| ☐ C Water cannot lose heat | ☐ D Water boils at 37 °C |

EM-LO3

Q3. Why does sweating cool the body?

- | | |
|--|---|
| ☐ A Sweat freezes on the skin | ☐ B Evaporating sweat absorbs heat from the skin |
| ☐ C Sweat condenses and releases heat | ☐ D Sweat has a low specific heat |

EM-LO4

Q4. On the heating curve of water, why is the plateau at 100 °C much longer than the plateau at 0 °C?

- ☐ A Steam has a higher specific heat than ice
- ☐ B ΔH_{vap} (2260 J/g) is much larger than ΔH_{fus} (334 J/g)
- ☐ C Water boils more slowly than ice melts because it is hotter
- ☐ D The temperature rises during boiling

EM-LO5

Q5. In an energy diagram, the products lie above the reactants. Which statement is correct?

- ☐ A The reaction is exothermic and ΔH is negative
- ☐ B The reaction is endothermic and ΔH is positive
- ☐ C The products have stronger bonds than the reactants
- ☐ D Heat is written on the product side

EM-LO6

Q6. Which statement about ATP is correct?

- ☐ A Converting ATP to ADP absorbs about 31 kJ/mol
- ☐ B Converting ATP to ADP releases about 31 kJ/mol, used to drive other reactions in cells
- ☐ C ATP releases more energy per mole than glucose
- ☐ D Converting ATP to ADP involves no bond breaking or forming

2 · Calculations

EM-LO1

Q7. A bottle of oral nutrition supplement provides 330 kcal. What is this in kJ? (1 kcal = 4.184 kJ)

Working:

Answer: _____ kJ

EM-LO2

Q8. A tourist's thermometer reads 102.2 °F. What is this in °C? Is it a fever?

Working:

Answer: _____ °C

EM-LO2

Q9. How much heat is needed to warm 500. g of IV fluid from 25.0 °C to 37.0 °C? Treat it as water ($c = 4.184 \text{ J/g}\cdot^\circ\text{C}$). Answer in kJ.

Working:

Answer: _____ kJ

EM-LO2

Q10. A 25.0 g metal sample absorbs 180. J and warms from 22.0 °C to 38.0 °C. Find its specific heat. (Then use the specific heat table to identify it.)

Working:

Answer: _____ $\text{J/g}\cdot^\circ\text{C}$

EM-LO3

Q11. A cold compress contains 15.0 g of ice at 0 °C. How much heat does it absorb from the skin as it melts completely? ($\Delta H_{\text{fus}} = 334 \text{ J/g}$). Answer in kJ.

Working:

Answer: _____ kJ

EM-LO3

Q12. 2.00 g of steam at 100 °C condenses on a hand. How much heat is released onto the skin by condensation alone? ($\Delta H_{\text{vap}} = 2260 \text{ J/g}$). Answer in kJ.

Working:

Answer: _____ kJ

EM-LO5

Q13. Estimate ΔH for burning methane: $\text{CH}_4 + 2 \text{O}_2 \rightarrow \text{CO}_2 + 2 \text{H}_2\text{O(g)}$. Bond energies (kJ/mol): C-H 418, O=O 498, C=O 795, O-H 460.

Working:

Answer: _____ kJ

EM-LO6

Q14. A patient with low blood sugar takes two glucose tablets containing 4.50 g of glucose each. How much energy can this glucose release when oxidised? (Glucose: 180.2 g/mol, releases 2800 kJ/mol.)

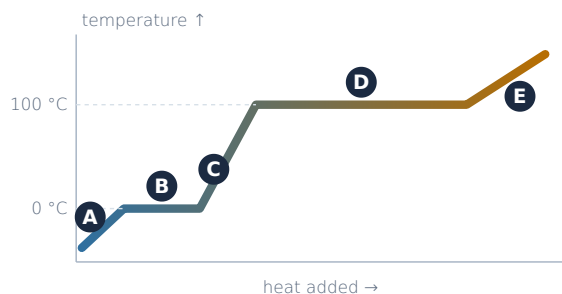
Working:

Answer: _____ kJ

3 · Label the heating curve

EM-LO4

Q15. Write the letter of each segment (A–E) next to the matching description.



- | | |
|-------|---|
| _____ | Water boiling at 100 °C: $q = m\Delta H_{\text{vap}}$ |
| _____ | Ice warming: $q = mc\Delta T$ with c of ice |
| _____ | Steam warming: $q = mc\Delta T$ with c of steam |
| _____ | Ice melting at 0 °C: $q = m\Delta H_{\text{fus}}$ |
| _____ | Water warming: $q = mc\Delta T$ with c of water |

4 · Pharmacy scenario

A hospital pharmacy prepares for a morning theatre list. Bags of saline are warmed from room temperature to body temperature before use, vaccines are sent to a clinic in a cold box packed with frozen ice packs, and a nurse asks for an instant cold pack for a patient's swollen wrist.

EM-LO2 (a) Calculate the heat needed to warm one 1.00 L bag of saline (take it as 1000 g of water) from 25.0 °C to 37.0 °C. Answer in kJ.

EM-LO3 · EM-LO4 (b) Why are frozen ice packs better than the same mass of cold water for keeping vaccines at 2–8 °C during transport?

EM-LO5 (c) The instant cold pack feels cold when squeezed. Is the process endothermic or exothermic? Explain using the words system and surroundings.

5 · Reflect

What was your 'muddiest point' in this lecture, the idea that is still unclear?

Explain the difference between heat and temperature in one or two sentences, as if to a friend.

6 · Your results

Outcome	Questions	Lecture part	My score
EM-LO1	Q1, Q7	Part 1	/ 2
EM-LO2	Q2, Q8, Q9, Q10	Part 2	/ 4
EM-LO3	Q3, Q11, Q12	Part 3	/ 3
EM-LO4	Q4, Q15 (5 labels)	Part 3	/ 6
EM-LO5	Q5, Q13	Part 4	/ 2
EM-LO6	Q6, Q14	Part 5	/ 2
Total			/ 19

The pharmacy scenario and reflection are not scored.

Worked answers

Q	Outcome	Answer	Working / why
Q1	EM-LO1	B	1 kcal = 4.184 kJ = 4184 J. 1 kJ and 1000 J are the same, and 100 cal is only 418.4 J.
Q2	EM-LO2	B	Water's specific heat (4.184 J/g·°C) is one of the highest. It absorbs or loses a lot of heat for a small temperature change, so the bath stays steady.
Q3	EM-LO3	B	Vaporisation is endothermic. The sweat takes its heat of vaporisation from the skin, so the skin cools.
Q4	EM-LO4	B	The length of a plateau is the heat needed for that phase change. Vaporising water needs about 7 times more heat per gram than melting ice.
Q5	EM-LO5	B	'Uphill' means the products have more energy, so energy must be absorbed: endothermic, ΔH positive. The reactants have the stronger bonds, and heat is written on the reactant side.
Q6	EM-LO6	B	ATP → ADP breaks an O–P bond and forms an O–P bond and two O–H bonds, releasing about 31 kJ/mol (7.5 kcal/mol). That is small compared with glucose (2800 kJ/mol) but enough to drive other reactions.
Q7	EM-LO1	1.38×10^3 kJ	$330 \text{ kcal} \times 4.184 \text{ kJ/kcal} = 1380.7 \text{ kJ} = \mathbf{1.38 \times 10^3 \text{ kJ}}$.
Q8	EM-LO2	39.0 °C	$^{\circ}\text{C} = (^{\circ}\text{F} - 32) \times 5/9 = (102.2 - 32) \times 5/9 = 70.2 \times 5/9 = \mathbf{39.0^{\circ}\text{C}}$. Yes: in adults, above 37.8 °C (100 °F) is a fever.
Q9	EM-LO2	25.1 kJ	$\Delta T = 37.0 - 25.0 = 12.0^{\circ}\text{C}$ $q = m \times c \times \Delta T = 500. \text{ g} \times 4.184 \text{ J/g}^{\circ}\text{C} \times 12.0^{\circ}\text{C} = 25\,104 \text{ J} = \mathbf{25.1 \text{ kJ}}$.
Q10	EM-LO2	0.450 J/g·°C	$\Delta T = 38.0 - 22.0 = 16.0^{\circ}\text{C}$ $c = q \div (m \times \Delta T) = 180. \div (25.0 \times 16.0) = \mathbf{0.450 \text{ J/g}^{\circ}\text{C}}$. This matches iron (0.452 J/g·°C).
Q11	EM-LO3	5.01 kJ	$q = m \times \Delta H_{\text{fus}} = 15.0 \text{ g} \times 334 \text{ J/g} = 5010 \text{ J} = \mathbf{5.01 \text{ kJ}}$. The temperature stays at 0 °C while it melts.
Q12	EM-LO3	4.52 kJ	$q = m \times \Delta H_{\text{vap}} = 2.00 \text{ g} \times 2260 \text{ J/g} = 4520 \text{ J} = \mathbf{4.52 \text{ kJ}}$, released. The condensed water then gives off even more heat as it cools, which is why steam burns are so severe.
Q13	EM-LO5	–762 kJ	Broken: $4 \text{ C-H} + 2 \text{ O=O} = 4(418) + 2(498) = 1672 + 996 = 2668 \text{ kJ}$ Formed: $2 \text{ C=O} + 4 \text{ O-H} = 2(795) + 4(460) = 1590 + 1840 = 3430 \text{ kJ}$ $\Delta H \approx 2668 - 3430 = \mathbf{-762 \text{ kJ}}$. Negative, so exothermic. (Average bond energies give an estimate; the measured value differs a little.)
Q14	EM-LO6	140 kJ	Mass = $2 \times 4.50 \text{ g} = 9.00 \text{ g}$ Moles = $9.00 \text{ g} \div 180.2 \text{ g/mol} = 0.0499 \text{ mol}$ Energy = $0.0499 \text{ mol} \times 2800 \text{ kJ/mol} = \mathbf{140 \text{ kJ}}$ (about 33 kcal).

Q15	EM-LO4	see right	A: Ice warming: $q = mc\Delta T$ with c of ice B: Ice melting at 0 °C: $q = m\Delta H_{\text{fus}}$ C: Water warming: $q = mc\Delta T$ with c of water D: Water boiling at 100 °C: $q = m\Delta H_{\text{vap}}$ E: Steam warming: $q = mc\Delta T$ with c of steam
Scenario (a)	EM-LO2		$q = m \times c \times \Delta T = 1000 \text{ g} \times 4.184 \text{ J/g}\cdot^{\circ}\text{C} \times 12.0^{\circ}\text{C} = 50\,208 \text{ J} =$ 50.2 kJ.
Scenario (b)	EM-LO3 · EM-LO4		As the ice melts it absorbs a lot of heat (334 J/g) while staying at 0 °C , because a phase change happens at constant temperature. The same mass of cold water only absorbs 4.184 J/g for each 1 °C rise, so it warms up much sooner.
Scenario (c)	EM-LO5		Endothermic. The system (the chemicals in the pack) absorbs heat from the surroundings (the wrist and the air), so the surroundings lose energy and feel cold.

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